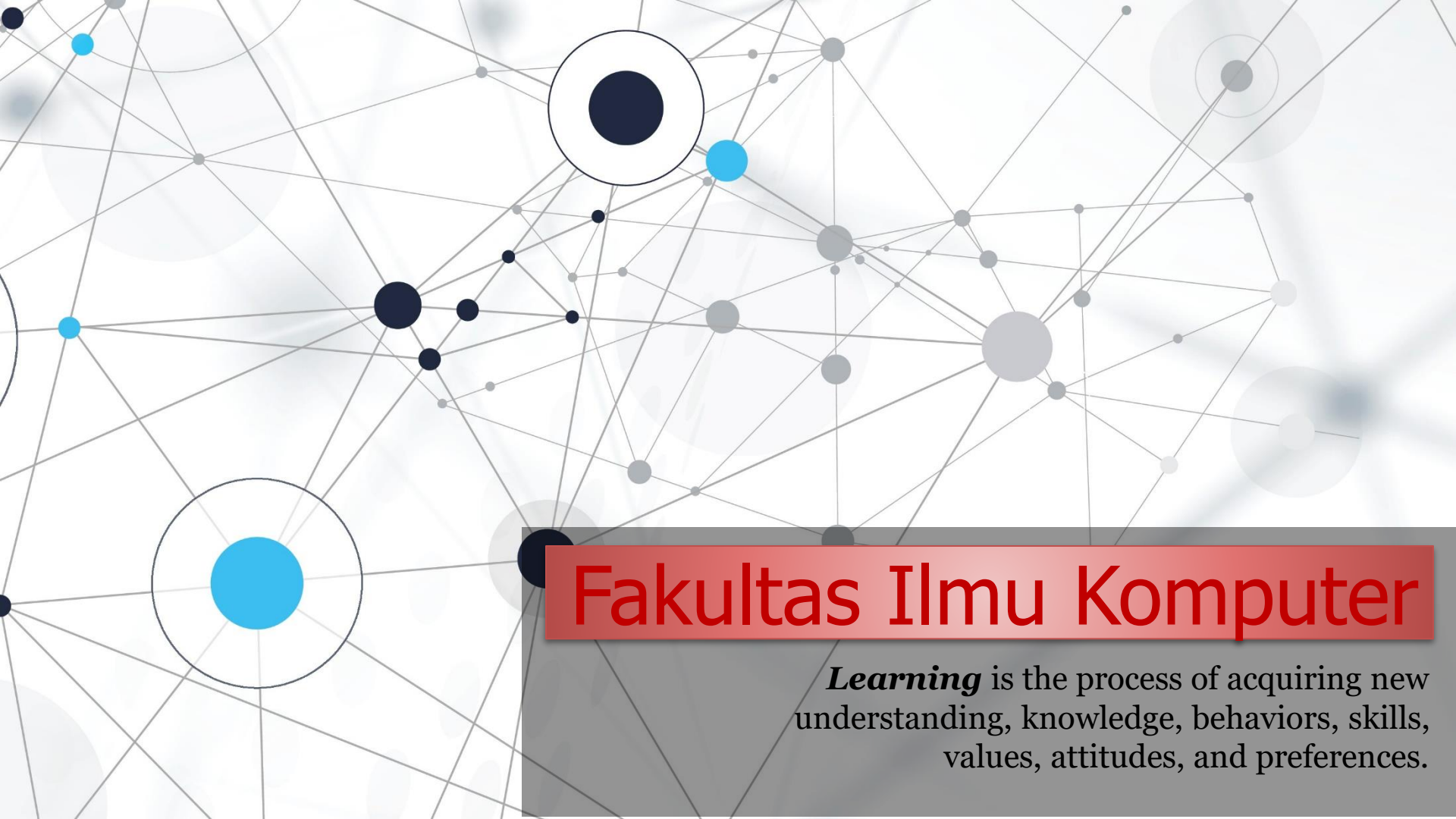


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

OOP in Python – Environment Setup



**Welcome to *the* Object Oriented
Programming *course***



Vibe coding
(or vibecoding) is an
approach to producing
software by using
artificial intelligence
(AI), where a person
describes a problem in a
few sentences as a
prompt to a large
language model (LLM)
tuned for coding.

-- Wikipedia

CODING vs VIBE CODING



- **Structure**
- **Logic**
- **Scalability**



- **Flow**
- **Speed**
- **Creativity**

Imagine building a software tool without writing a single line of code. No setup, no syntax errors, and no steep learning curve.

PYTHON FEATURES

01 GUI Programming

02 Easy to Understand

03 Free and Open Source

04 Object Oriented

05 Portable



06 Robust Standart Libraries

07 Scalable

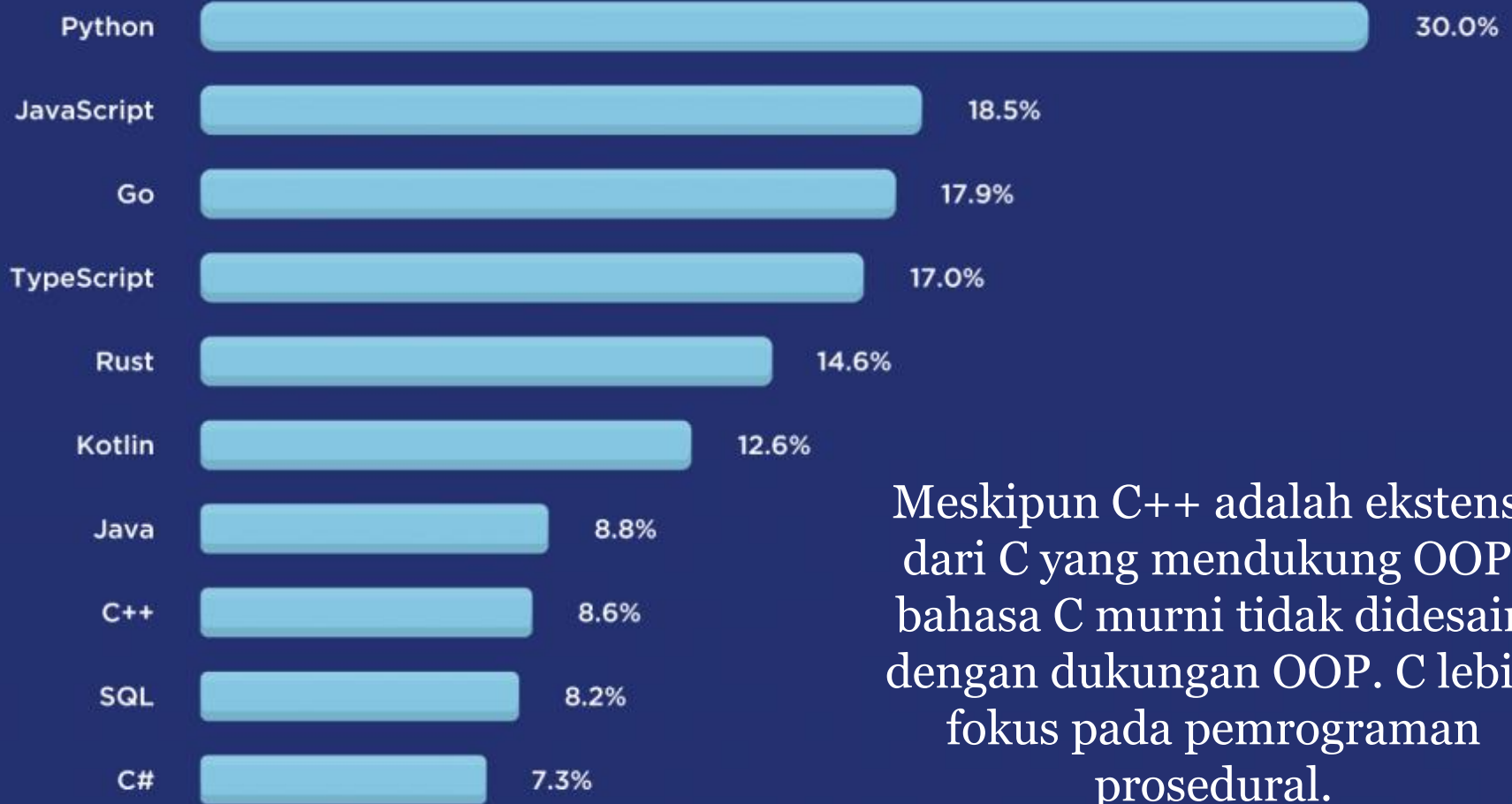
08 Supports Various Programming

09 Support for Interactive mode

10 Automatic Garbage Collection

Did you know that **most of the programming languages** extensively used in the computing world are object-oriented? With this help, **programmers** can create a lot of computer programs and a significant part of web content.

THE MOST WANTED TOP PROGRAMMING LANGUAGES



Meskipun C++ adalah ekstensi dari C yang mendukung OOP, bahasa C murni tidak didesain dengan dukungan OOP. C lebih fokus pada pemrograman prosedural.

The list of high-level programming languages includes Java, Python, C#, Ruby, Swift (Apple, untuk pengembangan aplikasi iOS dan macOS), Kotlin (berjalan di atas Java Virtual Machine), Javascript, PHP, Scala, etc. **Which programming languages are not object-oriented?** These are C, COBOL, Fortran, Forth, and Pascal.

Structure of Object-Oriented Programming

Menurut kalian, mengapa banyak bahasa pemrograman modern mendukung paradigma pemrograman berorientasi objek (OOP)? Bahkan, ada banyak bahasa pemrograman yang secara khusus didesain atau memiliki dukungan kuat untuk OOP pasti ada alasannya.



The widespread support for OOP in modern programming languages is based on the benefits it brings to the maintenance, extension, and development of software projects.

The OOP paradigm promotes more **understandable** and **structured code**. Concepts such as classes, objects, and inter-object relations allow developers to structure and understand code better.

ARTIFICIAL INTELLIGENCE

Early artificial intelligence stirs excitement.



MACHINE LEARNING

Machine learning begins to flourish.



DEEP LEARNING

Deep learning breakthroughs drive AI boom.



1950's 1960's 1970's 1980's 1990's 2000's 2010's

Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

AI is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.



Kemampuan AI di Tahun 2016, di train menggunakan bahasa pemrograman apa?

Bagaimana cara memilih bahasa pemrograman yang paling bagus untuk mempelajari Pemrograman Berorientasi Objek (OOP) ???



Remember, to choose a programming language is a personal decision, and there is no right or wrong choice. The most important thing is to choose a language that suits your goals and interests in software development using an object-oriented approach.



[1] Learning Objectives (Tujuan Pembelajaran)

Tentukan tujuan dalam mempelajari OOP !!!

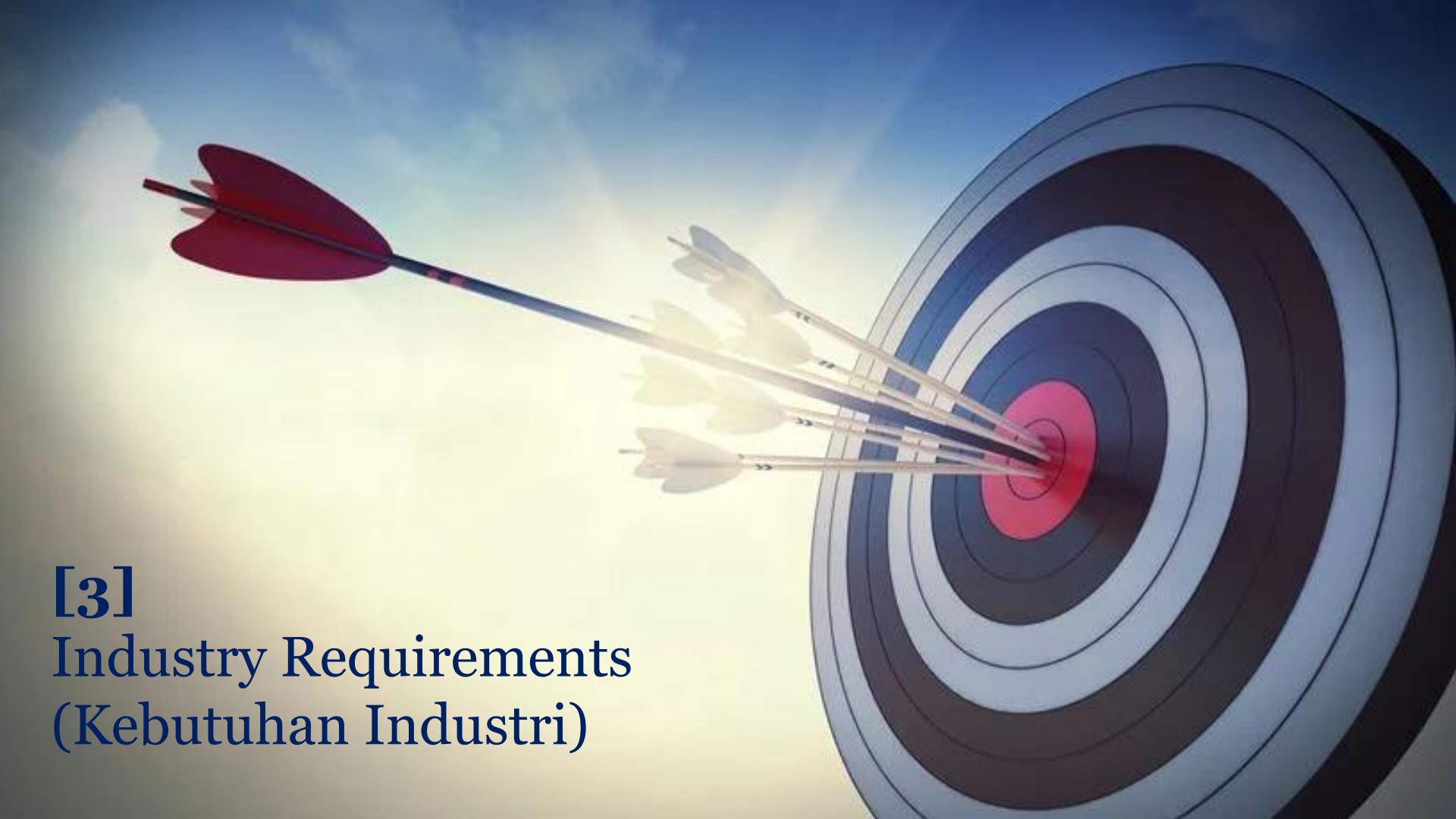
Do you want to create web applications, mobile applications, desktop software, or other projects? The choice of programming language can depend on the type of project you want to work on.



[2]

Bahasa pemrograman yang populer dan memiliki banyak sumber daya

Choose a programming language that is popular and has many learning resources. Languages like Java, Python, and C# are very popular for object-oriented programming and have large communities.



[3]

Industry Requirements (Kebutuhan Industri)

If you have specific plans, such as Android app development (using Kotlin or Java) or game development (using C# with Unity), consider a language that is commonly used in that space.



1 – Python Environment

Prerequisites & Toolkits

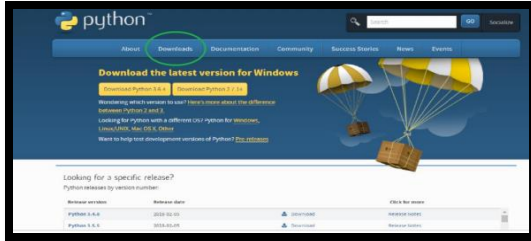
Before proceed with learning further on Python, we suggest you to check whether the following prerequisites are met:

- ❖ Latest version of Python is installed on your computer.
- ❖ An IDE or text editor is installed.
- ❖ Have basic familiarity to write and debug in Python:
 - i. Able to write and run Python programs.**
 - ii. Debug programs and diagnose errors.**
 - iii. Work with basic data types.**
 - iv. Write for loops, while loops, and if statements**
 - v. Code functions**

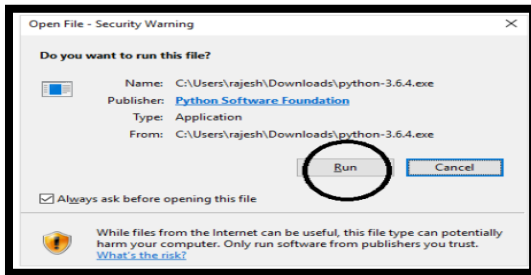
If you don't have any programming language experience, you can find lots of beginner tutorials in Python:

- 1) <https://www.w3schools.com/python/>
- 2) <https://www.pythontutorial.net/>
- 3) <https://www.tutorialspoint.com/python/index.htm>
- 4) <https://www.petanikode.com/tutorial/python/>
- 5) <https://www.programiz.com/python-programming>

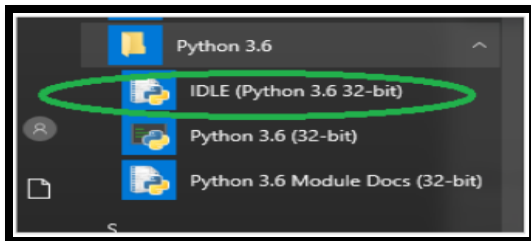
Installing Python



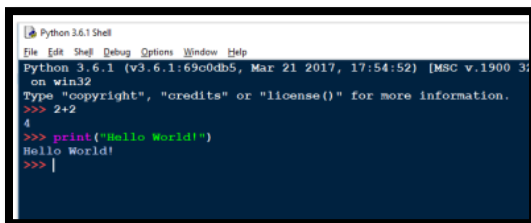
- ❖ **Step 1:** Go to the official Python website <https://www.Python.org/>, click on the Downloads menu and choose the latest or any stable version of your choice.



- ❖ **Step 2:** Save the Python installer exe file that you're downloading and once you have downloaded it, open it. Click on Run and choose Next option by default and finish the installation.



- ❖ **Step 3:** After you have installed, you should now see the Python menu as shown in the image below. Start the program by choosing IDLE (Python GUI).



- ❖ This will start the Python shell. Type in simple commands to check the installation.

Choosing *an* IDE

An IDE (Integrated Development Environment) is a **text editor** geared towards software development. You will have to install an IDE to control the flow of your programming and to group projects together when working on Python. Here are some of IDEs available online. You can choose one at your convenience.

- 1) Pycharm IDE
- 2) Komodo IDE
- 3) Eric Python IDE
- 4) Notepad++
- 5) **Spyder IDE**
- 6) **Note:** Eclipse IDE is mostly used in Java, however it has a Python plugin.

The slide features a white background with green leaves framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub, with some showing signs of aging or damage. The text is centered in a black serif font.

2 - Integrated Development Environment

Pycharm

Pycharm, the cross-platform IDE is one of the most popular IDE currently available. It provides coding assistance and analysis with code completion, project and code navigation, integrated unit testing, version control integration, debugging and much more.

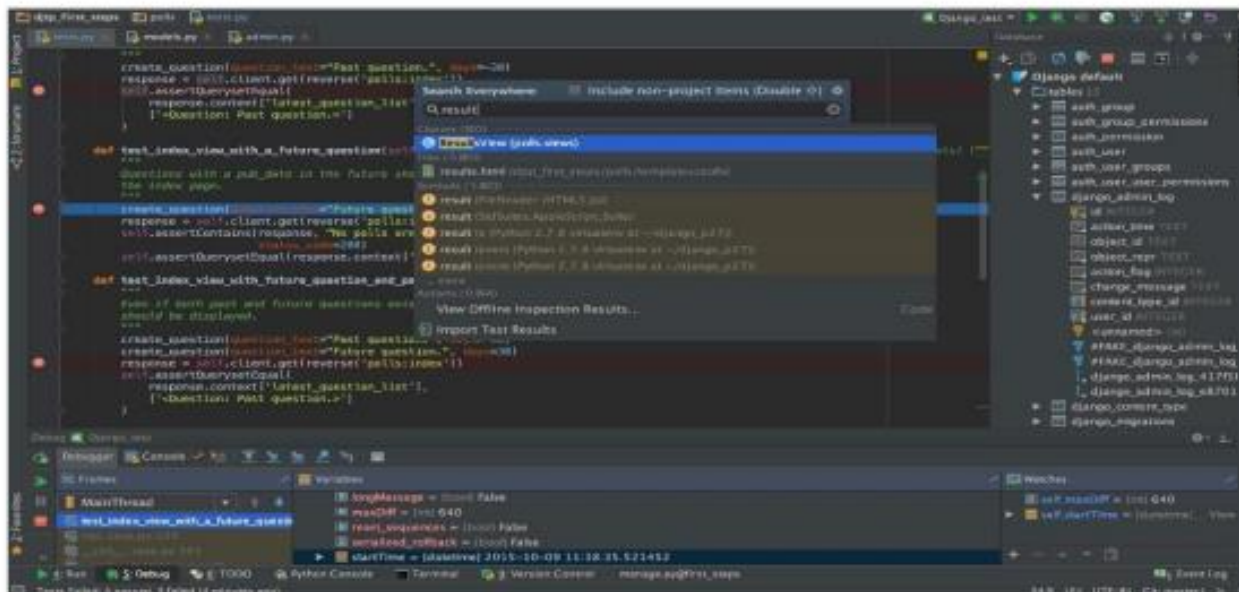


Download link

<https://www.jetbrains.com/pycharm/download/>

Languages Supported: Python, HTML, CSS, JavaScript, Coffee Script, TypeScript, Cython, AngularJS, Node.js, template languages.

Screenshot



Komodo IDE

It is a polyglot IDE which supports 100+ languages and basically for dynamic languages such as Python, PHP and Ruby. It is a commercial IDE available for 21 days free trial with full functionality. ActiveState is the software company managing the development of the Komodo IDE. It also offers a trimmed version of Komodo known as Komodo Edit for simple programming tasks.



This IDE contains all kinds of features from most basic to advanced level. If you are a student or a freelancer, then you can buy it almost half of the actual price. However, it's completely free for teachers and professors from recognized institutions and universities.

It got all the features you need for web and mobile development, including support for all your languages and frameworks.

Download link

The download links for Komodo Edit(free version) and Komodo IDE(paid version) are as given here:

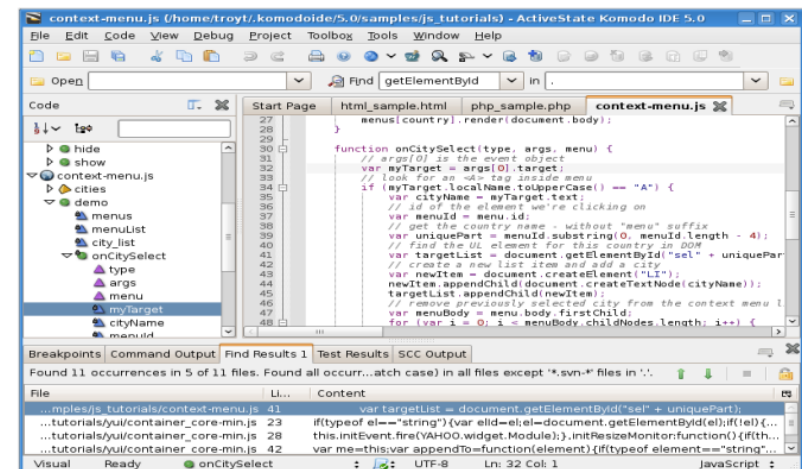
Komodo Edit (free)

<https://www.activestate.com/komodo-edit>

Komodo IDE (paid)

<https://www.activestate.com/komodo-ide/downloads/ide>

Screenshot



Eric Python IDE

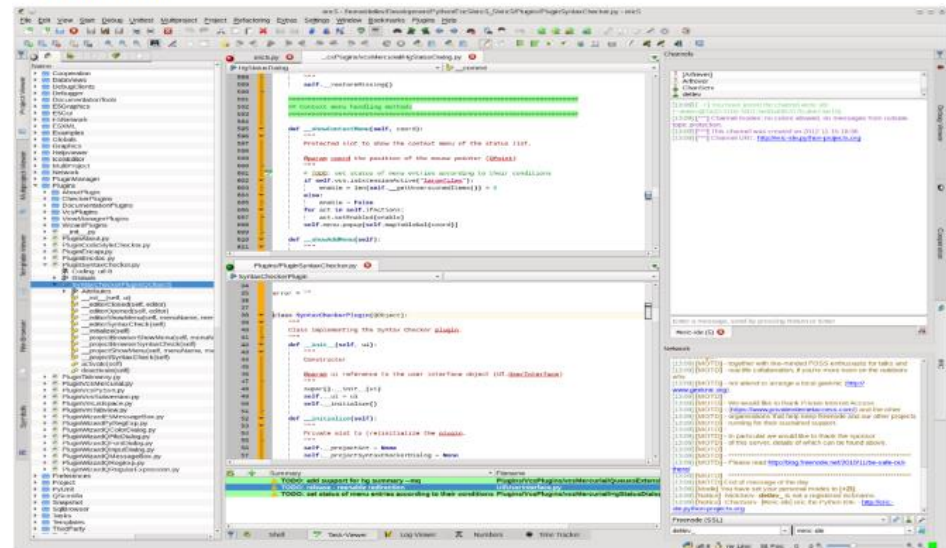
It is an open-source IDE for Python and Ruby. Eric is a full featured editor and IDE, written in Python. It is based on the cross platform Qt GUI toolkit, integrating the highly flexible Scintilla editor control. The IDE is very much configurable and one can choose what to use and what not. You can download Eric IDE from below link:
<https://eric-ide.Python-projects.org/eric-download.html>



Why to Choose

- Great indentation, error highlighting.
- Code assistance
- Code completion
- Code cleanup with Pylint
- Quick search
- Integrated Python debugger.

Screenshot



Choosing a Text Editor

For tasks such as learning to code with Python or Arduino, or when working on a quick script in shell script to help you automate some tasks a simple and light weight code-centric text editor will do. Also many text editors offer features such as syntax highlighting and in-program script execution, similar to IDEs. Some of the text editors are given here:

- 1) Atom
- 2) Sublime Text
- 3) Notepad++

Atom Text Editor

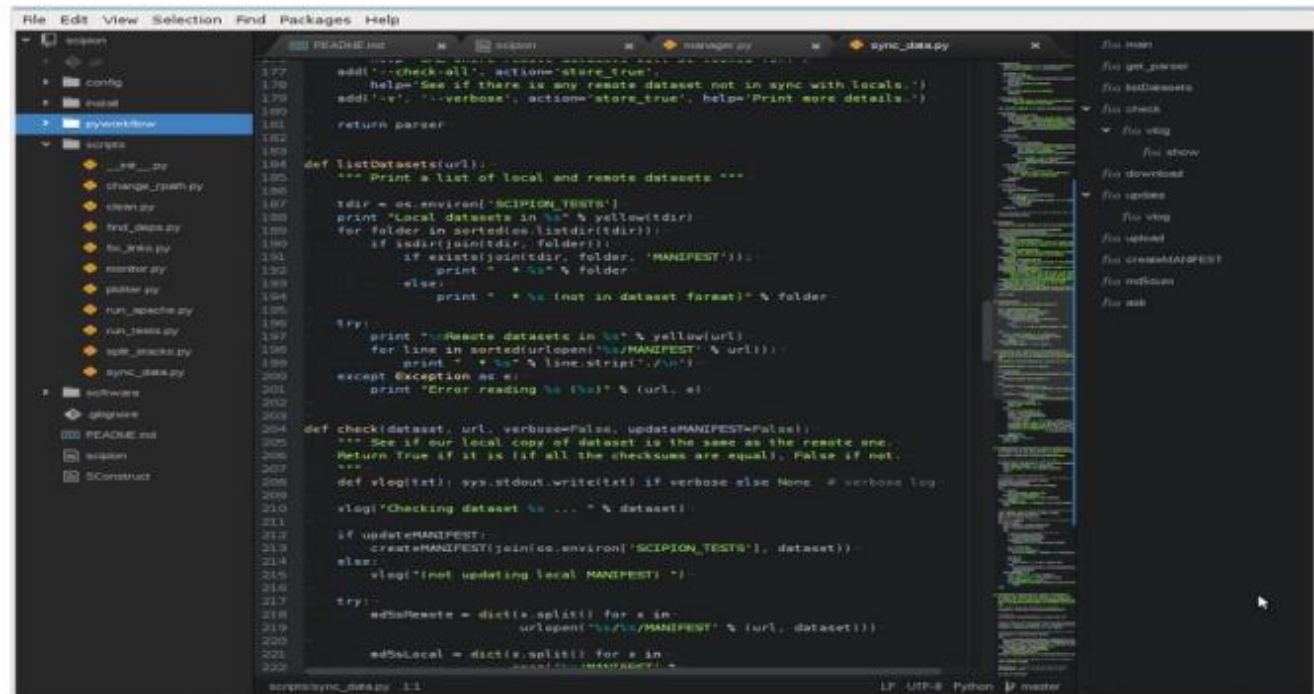
Atom is a hackable text editor built by the team of GitHub. It is a free and open source text and code editor which means that all the code is available for you to read, modify for your own use and even contribute improvements. It is a cross-platform text editor compatible for macOS, Linux, and Microsoft Windows with support for plug-ins written in Node.js and embedded Git Control.



Download link

<https://atom.io/>

Screenshot



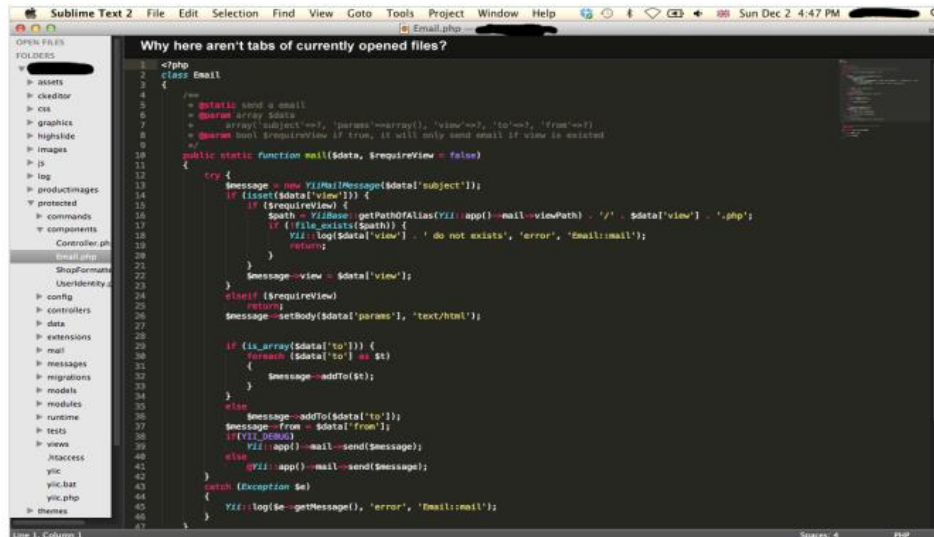
Sublime Text Editor

Sublime text is a proprietary software and it offers you a free trial version to test it before you purchase it. According to [Stackoverflow's 2018 developer survey](#), it's the fourth most popular Development Environment.



Some of the advantages it provides is its incredible speed, ease of use and community support. It also supports many programming languages and mark-up languages, and functions can be added by users with plugins, typically community-built and maintained under free-software licenses.

Screenshot



Language supported

- Python, Ruby, JavaScript etc.

Notepad++

It's a free source code editor and Notepad replacement that supports several languages from Assembly to XML and including Python. Running in the MS windows environment, its use is governed by GPL license. In addition to syntax highlighting, Notepad++ has some features that are particularly useful to coders.



Screenshot

```
1  #include <GPL.h>
2  #include <free_software.h>
3
4  void notepad4ever ()
5  {
6      while (true)
7      {
8          Notepad++;
9      }
10 }
11
```

Language Supported

Almost every language (60+ languages) like Python, C, C++, C#, Java etc.

Key Features

- Syntax highlighting and syntax folding
- PCRE (Perl Compatible Regular Expression) Search/Replace.
- Entirely customizable GUI
- Auto completion
- Tabbed editing
- Multi-View
- Multi-Language environment
- Launchable with different arguments.



Exercise *for* Students

Programming paradigms *in* Python

There are three programming paradigms that are supported by Python:

1. Procedural programming
2. Object-Oriented Programming
3. Functional programming

[1] Procedural Programming

The image shows the Spyder Python IDE interface. The main editor window displays a Python script named `Procedural.py` with the following content:

```
1 # -*- coding: utf-8 -*-
2 """
3 @Created : Tue Jan 17 06:52:31 2023
4 @author : SWT
5 @desc : Simple Procedural Programming
6 """
7
8 # variables
9 species = "bird";
10 parrot_name = "parrot";
11 sparrow_name = "sparrow";
12 parrot_age = "10";
13 sparrow_age = "15";
14
15 # view variable parrot (burung beo) & sparrow (burung gereja)
16 print("1. parrot is a {}".format(species))
17 print("2. sparrow is also a {}".format(species))
18 print("3. {} is {} years old".format(parrot_name, parrot_age))
19 print("4. {} is {} years old".format(sparrow_name, sparrow_age))
20
21
```

The IPython console on the right shows the execution of the script:

```
Python 3.9.7 (default, Sep 16 2021, 16:59:28) [MSC v. 1916 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

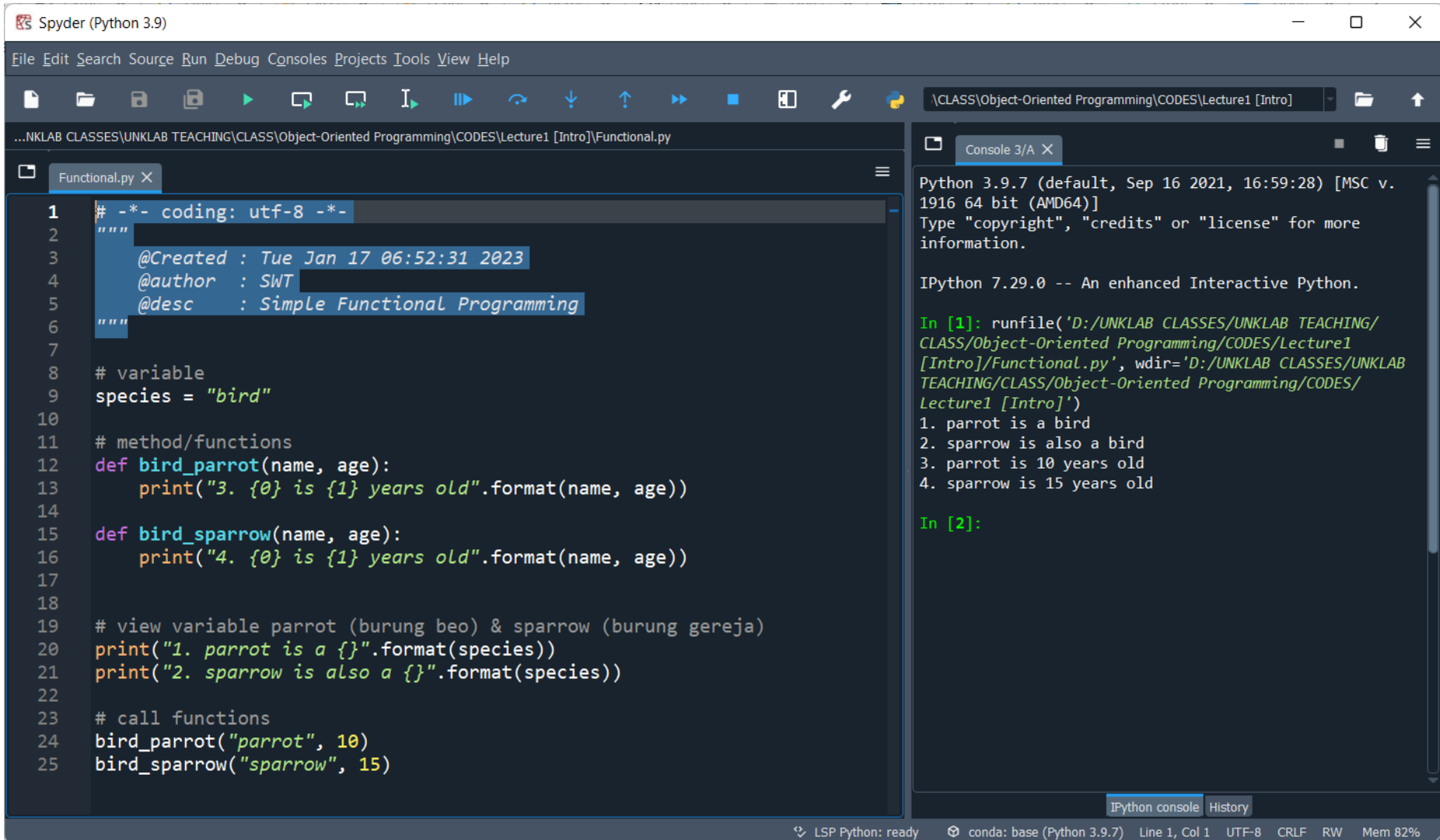
IPython 7.29.0 -- An enhanced Interactive Python.

In [1]: runfile('D:/UNKLAB CLASSES/UNKLAB TEACHING/CLASS/Object-Oriented Programming/CODES/Lecture1 [Intro]/Procedural.py', wdir='D:/UNKLAB CLASSES/UNKLAB TEACHING/CLASS/Object-Oriented Programming/CODES/Lecture1 [Intro]')
1. parrot is a bird
2. sparrow is also a bird
3. parrot is 10 years old
4. sparrow is 15 years old

In [2]:
```

The status bar at the bottom indicates: LSP Python: ready, conda: base (Python 3.9.7), Line 1, Col 1, UTF-8, CRLF, RW, Mem 81%.

[2] Functional Programming



The image shows the Spyder Python IDE interface. The main editor window displays a Python script named `Functional.py` with the following content:

```
1  # -*- coding: utf-8 -*-
2  """
3      @Created : Tue Jan 17 06:52:31 2023
4      @author  : SWT
5      @desc   : Simple Functional Programming
6  """
7
8  # variable
9  species = "bird"
10
11 # method/functions
12 def bird_parrot(name, age):
13     print("3. {0} is {1} years old".format(name, age))
14
15 def bird_sparrow(name, age):
16     print("4. {0} is {1} years old".format(name, age))
17
18
19 # view variable parrot (burung beo) & sparrow (burung gereja)
20 print("1. parrot is a {}".format(species))
21 print("2. sparrow is also a {}".format(species))
22
23 # call functions
24 bird_parrot("parrot", 10)
25 bird_sparrow("sparrow", 15)
```

The IPython console on the right shows the execution output:

```
Python 3.9.7 (default, Sep 16 2021, 16:59:28) [MSC v. 1916 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

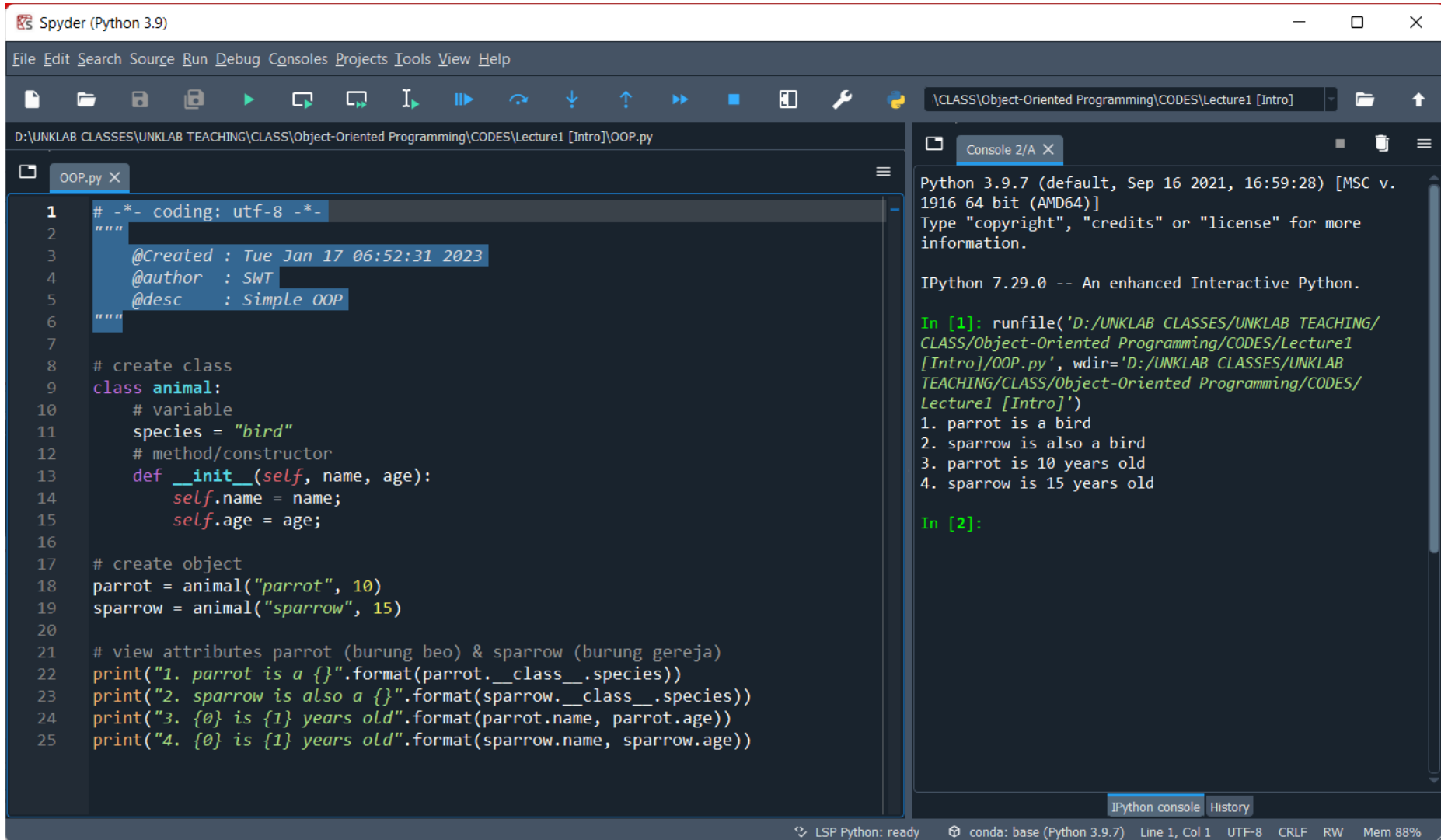
IPython 7.29.0 -- An enhanced Interactive Python.

In [1]: runfile('D:/UNKLAB CLASSES/UNKLAB TEACHING/CLASS/Object-Oriented Programming/CODES/Lecture1 [Intro]/Functional.py', wdir='D:/UNKLAB CLASSES/UNKLAB TEACHING/CLASS/Object-Oriented Programming/CODES/Lecture1 [Intro]')
1. parrot is a bird
2. sparrow is also a bird
3. parrot is 10 years old
4. sparrow is 15 years old

In [2]:
```

The status bar at the bottom indicates: LSP Python: ready, conda: base (Python 3.9.7), Line 1, Col 1, UTF-8, CRLF, RW, Mem 82%.

[3] Object-Oriented Programming



The image shows the Spyder Python IDE interface. The main editor window displays a Python script named `OOP.py` with the following content:

```
1 # -*- coding: utf-8 -*-
2 """
3 @Created : Tue Jan 17 06:52:31 2023
4 @author : SWT
5 @desc : Simple OOP
6 """
7
8 # create class
9 class animal:
10     # variable
11     species = "bird"
12     # method/constructor
13     def __init__(self, name, age):
14         self.name = name;
15         self.age = age;
16
17 # create object
18 parrot = animal("parrot", 10)
19 sparrow = animal("sparrow", 15)
20
21 # view attributes parrot (burung beo) & sparrow (burung gereja)
22 print("1. parrot is a {}".format(parrot.__class__.species))
23 print("2. sparrow is also a {}".format(sparrow.__class__.species))
24 print("3. {} is {} years old".format(parrot.name, parrot.age))
25 print("4. {} is {} years old".format(sparrow.name, sparrow.age))
```

The IPython console on the right shows the output of the script:

```
Python 3.9.7 (default, Sep 16 2021, 16:59:28) [MSC v. 1916 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

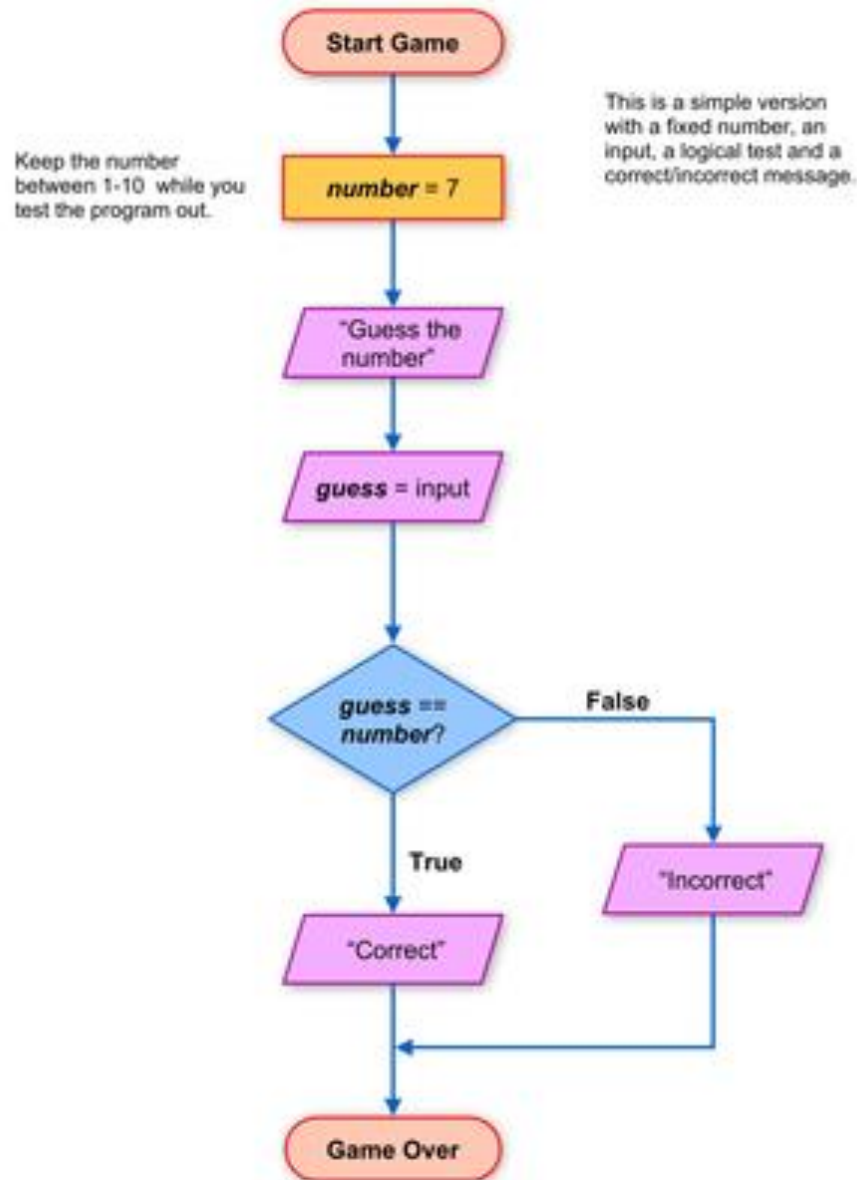
IPython 7.29.0 -- An enhanced Interactive Python.

In [1]: runfile('D:/UNKLAB CLASSES/UNKLAB TEACHING/CLASS/Object-Oriented Programming/CODES/Lecture1 [Intro]/OOP.py', wdir='D:/UNKLAB CLASSES/UNKLAB TEACHING/CLASS/Object-Oriented Programming/CODES/Lecture1 [Intro]')
1. parrot is a bird
2. sparrow is also a bird
3. parrot is 10 years old
4. sparrow is 15 years old

In [2]:
```

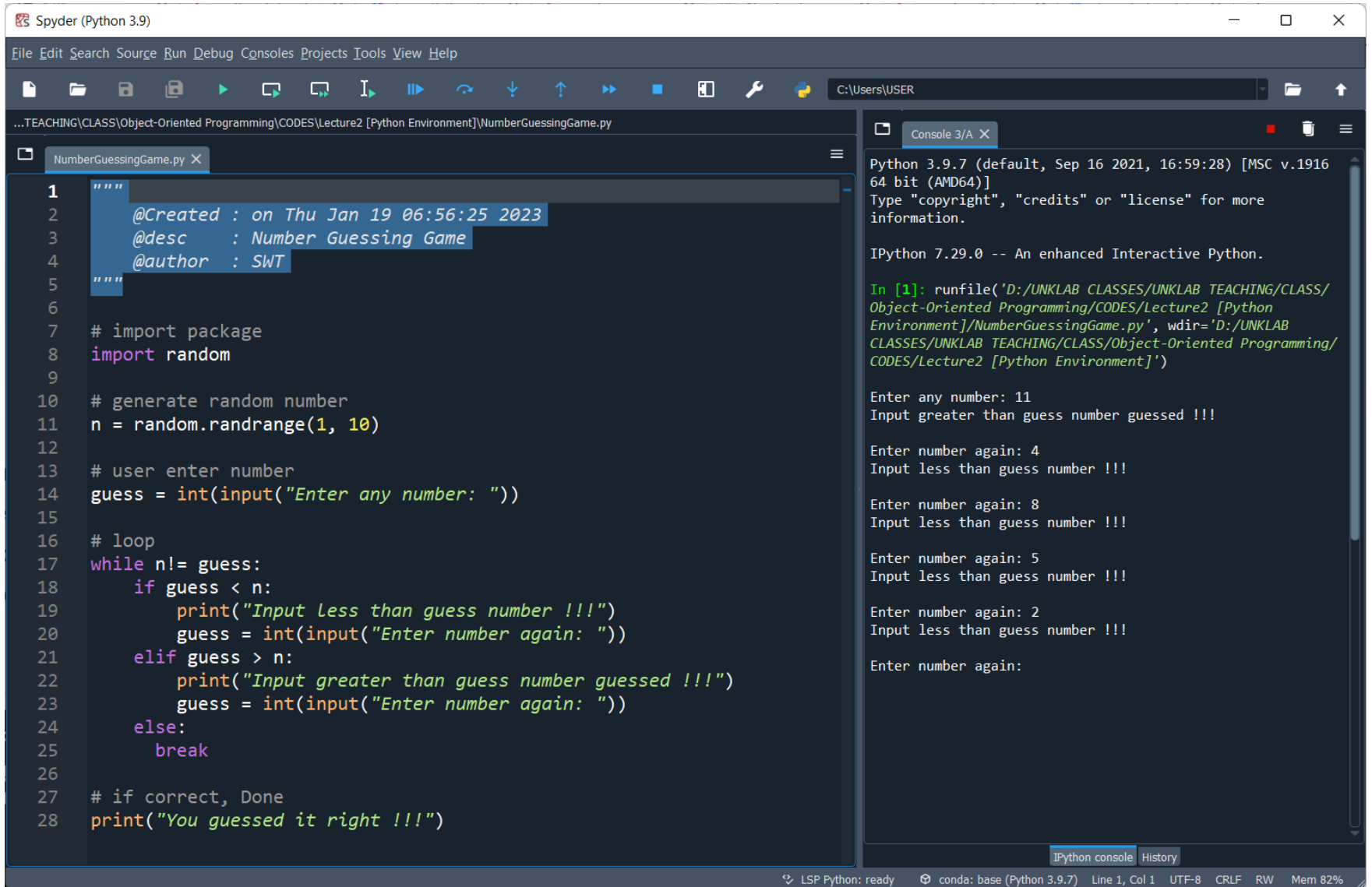
The status bar at the bottom indicates: LSP Python: ready, conda: base (Python 3.9.7), Line 1, Col 1, UTF-8, CRLF, RW, Mem 88%.

1 Number Guessing Game



Spyder, CMD & Google Colab

Spyder IDE



The screenshot displays the Spyder IDE (Python 3.9) interface. The main editor window shows a Python script named `NumberGuessingGame.py` with the following code:

```
1 """
2     @Created : on Thu Jan 19 06:56:25 2023
3     @desc    : Number Guessing Game
4     @author  : SWT
5 """
6
7 # import package
8 import random
9
10 # generate random number
11 n = random.randrange(1, 10)
12
13 # user enter number
14 guess = int(input("Enter any number: "))
15
16 # loop
17 while n!= guess:
18     if guess < n:
19         print("Input less than guess number !!!")
20         guess = int(input("Enter number again: "))
21     elif guess > n:
22         print("Input greater than guess number guessed !!!")
23         guess = int(input("Enter number again: "))
24     else:
25         break
26
27 # if correct, Done
28 print("You guessed it right !!!")
```

The right-hand pane shows the IPython console output:

```
Python 3.9.7 (default, Sep 16 2021, 16:59:28) [MSC v.1916
64 bit (AMD64)]
Type "copyright", "credits" or "license" for more
information.

IPython 7.29.0 -- An enhanced Interactive Python.

In [1]: runfile('D:/UNKLAB CLASSES/UNKLAB TEACHING/CLASS/
Object-Oriented Programming/CODES/Lecture2 [Python
Environment]/NumberGuessingGame.py', wdir='D:/UNKLAB
CLASSES/UNKLAB TEACHING/CLASS/Object-Oriented Programming/
CODES/Lecture2 [Python Environment]')

Enter any number: 11
Input greater than guess number guessed !!!

Enter number again: 4
Input less than guess number !!!

Enter number again: 8
Input less than guess number !!!

Enter number again: 5
Input less than guess number !!!

Enter number again: 2
Input less than guess number !!!

Enter number again:
```

The status bar at the bottom indicates: LSP Python: ready, conda: base (Python 3.9.7), Line 1, Col 1, UTF-8, CRLF, RW, Mem 82%.

Buatlah versi dari number guessing game (permainan tebak angka) dengan pendekatan functional programming?

Pendekatan ini menggunakan fungsi untuk membagi tugas-tugas dalam program dan memisahkan logika permainan ke dalam fungsi yang berbeda sehingga memberikan struktur kode yang lebih bersih.



Google Colaboratory (Colab)



google colab



All



Images



Videos



Books



News

More

Tools

About 25,300,000 results (0.39 seconds)

 <https://colab.research.google.com>

Welcome To Colaboratory - Colaboratory - Google

Colab notebooks allow you to combine executable code and rich text in a single document, along with images, HTML, LaTeX and more. When you create your own **Colab** ...

Colab

Anda dapat mengimpor data sendiri ke notebook Colab dari ...

Pro

Compute units expire after 90 days. Purchase more as you need ...

Filter notebooks

Overview of Colaboratory - Markdown Guide - Google Drive

Connect

Colab is a Python development environment that runs in the ...

[More results from google.com »](#)

Google Colaboratory (Colab)

The screenshot shows the Google Colaboratory web interface. The browser tab is titled 'Welcome To Colaboratory - Colab' and the address bar shows 'colab.research.google.com'. The interface has a top navigation bar with tabs: 'Examples', 'Recent' (selected), 'Google Drive', 'GitHub', and 'Upload'. Below the 'Recent' tab, there is a 'Filter notebooks' input field and a table of recent notebooks. The table has columns for 'Title', 'Last opened', 'First opened', and action icons. The notebooks listed are:

Title	Last opened	First opened	Actions
Download [EPub] Full Stack Web Development For Beginners Learn Eco...	January 17	January 17	Download, Open
Untitled21.ipynb	Nov 30, 2022	Nov 30, 2022	Download, Open
Lecture 23 (14 Nov 2022).ipynb	Nov 21, 2022	Nov 21, 2022	Download, Open
Untitled20.ipynb	Nov 21, 2022	Nov 21, 2022	Download, Open

At the bottom right of the interface, there are buttons for 'New notebook' and 'Cancel'. The status bar at the bottom shows the URL 'https://colab.research.google.com/#create=true'.

Google Colaboratory (Colab)

The screenshot displays the Google Colaboratory (Colab) web interface. The browser address bar shows the URL: `colab.research.google.com/drive/14XNaG2rsaBOWFDzsrHErNv35Cksjs5y7`. The Colab logo and the file name "NumberGuessingGame.ipynb" are visible in the top left. The top right contains links for "Comment", "Share", and a settings icon. Below the top bar, there are tabs for "Code" and "Text", and a status bar showing "RAM" and "Disk" usage. The main area contains a code cell with the following Python code:

```
# import package
import random

# generate random number
n = random.randrange(1, 10)

# user enter number
guess = int(input("Enter any number: "))

# loop
while n!= guess:
    if guess < n:
        print("Input less than guess number !!!")
        guess = int(input("Enter number again: "))
    elif guess > n:
        print("Input greater than guess number guessed !!!")
        guess = int(input("Enter number again: "))
    else:
        break

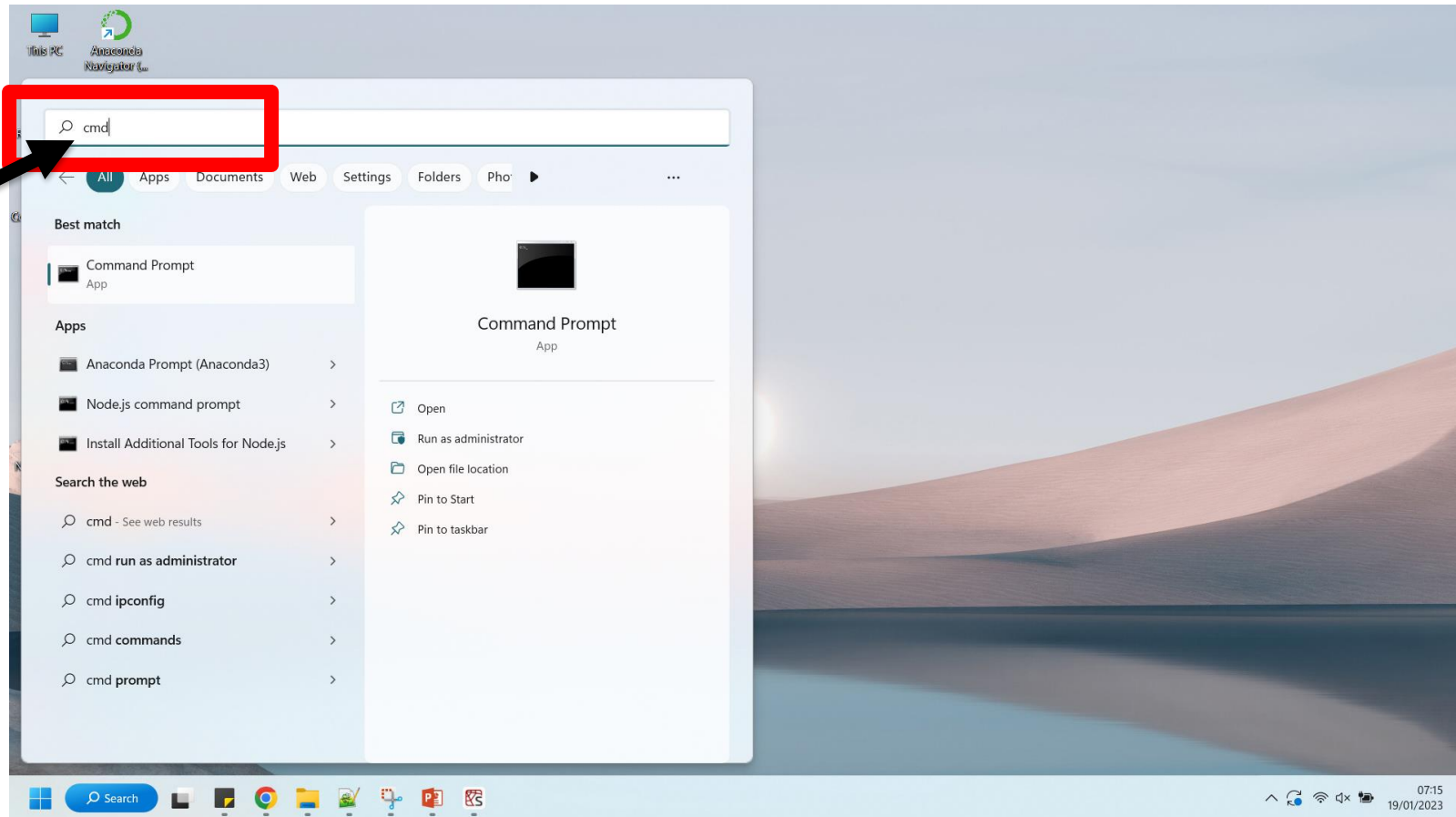
# if correct, Done
print("You guessed it right !!!")
```

Below the code cell, the output is shown, including the prompt "Enter any number: 12" and the feedback messages "Input greater than guess number guessed !!!", "Enter number again: 3", "Input less than guess number !!!", and "Enter number again:". A text input field is visible for the final prompt.

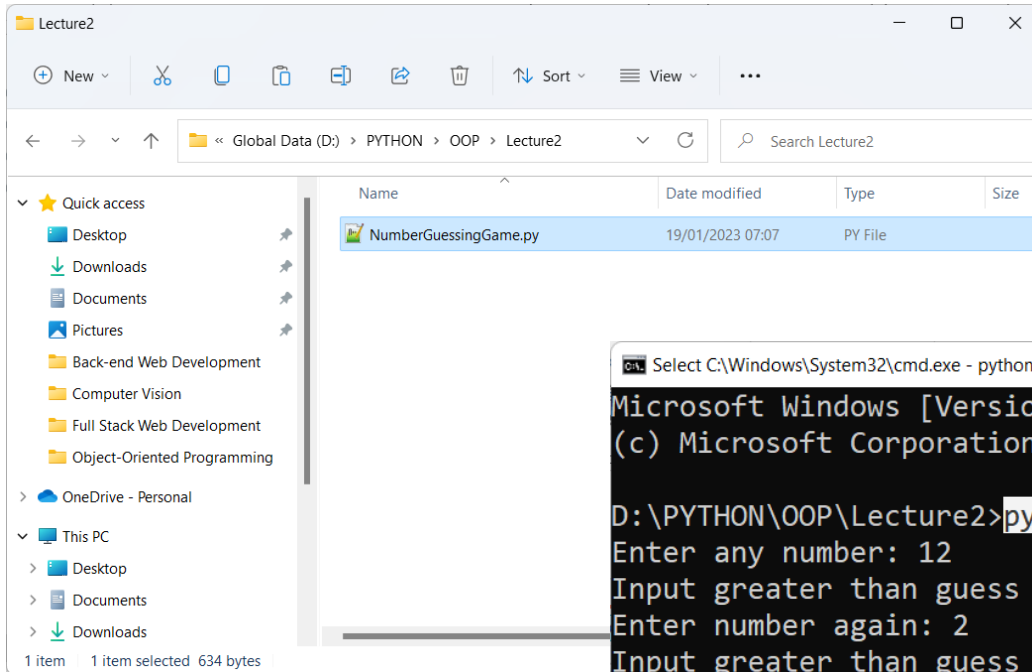
The bottom status bar indicates the execution progress: "Executing (31s) Cell > raw_input() > _input_request() > select()".

Command Prompt (Windows)

Search here !!!



Command Prompt (CMD)



Write on console:

> python NumberGuessingGame.py

The screenshot shows a Windows Command Prompt window titled 'Select C:\Windows\System32\cmd.exe - python NumberGuessingGame.py'. The prompt shows the command 'python NumberGuessingGame.py' being executed. The output of the script is as follows:

```
Microsoft Windows [Version 10.0.22000.1455]
(c) Microsoft Corporation. All rights reserved.

D:\PYTHON\OOP\Lecture2>python NumberGuessingGame.py
Enter any number: 12
Input greater than guess number guessed !!!
Enter number again: 2
Input greater than guess number guessed !!!
Enter number again: 3
Input greater than guess number guessed !!!
Enter number again: 5
Input greater than guess number guessed !!!
Enter number again: 6
Input greater than guess number guessed !!!
Enter number again: 5
Input greater than guess number guessed !!!
Enter number again: 4
Input greater than guess number guessed !!!
Enter number again: 
```

Test your understanding

The following are some questions to reinforce your understanding of the concepts we've covered so far.



Question #1

Students dapat menjelaskan dengan singkat, apa yang di maksud dengan Python IDE (Integrated Development Environment)? Sebutkan jenis-jenis python IDE yang kamu ketahui? Sebutkan beberapa text editor untuk Python yang kamu ketahui?

Question #2

Students dapat menjelaskan dengan singkat, kapan kita harus menggunakan Spyder IDE dan kapan kita harus menggunakan Google colab?

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

